A Study on Choppy Pulse Image Classification Using Multi-Scale Feature Fusion with a Light ConvNeXt-style Architecture*

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Abstract

Choppy pulse (CP) is one of the pulse patterns with important diagnostic value in traditional Chinese medicine pulse diagnosis. However, its assessment still depends

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heavily on the clinical experience of physicians, and therefore objective and standardized identification methods are still needed. In recent years, deep learning has been widely applied to medical image classification tasks. However, studies on CP waveform images have rarely investigated the complementary value of different field-of-view scales in a systematic manner. Therefore, this study proposes a CP image classification method based on multi-scale feature fusion with a Light ConvNeXt-style architecture. The purpose of this study is to compare the classification performance of single-scale models and multi-scale fusion models in the CP classification task, and to further examine the role and complementary effect of waveform information from different scales. The dataset was constructed from 6-second pulse wave images measured by a pulse diagnosis instrument in 56 CP cases confirmed by traditional Chinese medicine physicians. A single pulse waveform was first extracted from each 6-second recording. Then, multi-size images were cropped with the highest waveform peak as the center to establish a binary dataset of CP and Non-Choppy Pulse (NCP). The data were divided by patient-wise split into training, validation, and test sets in order to reduce the risk of data leakage. This study focused on three representative scales, namely 33×33, 61×61, and 105×105, which correspond to local detail, main contour, and global context, respectively. Six fixed experiments were designed, including three single-scale models, two dual-scale fusion models, and one triple-scale fusion model. The results show that multi-scale fusion models can integrate waveform features from different field-of-view levels and can improve the classification performance of CP images. Overall, this study suggests that CP images contain multi-level information, and that multi-scale feature fusion can provide a more complete basis for classification. These findings may serve as a useful reference for the development of intelligent assistance systems for traditional Chinese medicine pulse diagnosis and may further improve their completeness and practical value.

1 Introduction

Traditional Chinese Medicine (TCM) pulse diagnosis is one of the important foundations of clinical syndrome differentiation. However, its interpretation has long relied on physicians' clinical experience and subjective tactile perception, which still imposes clear limitations on standardization, objectification, and reproducibility. Previous studies have attempted to promote the quantification of pulse diagnosis through artificial intelligence models, classification index construction, and pulse waveform imaging methods, showing that pulse information can be transformed into more consistent quantitative descriptions and can further serve as a technical foundation for automated recognition (Tang *et al.* (2012), Shu and Sun (2007), Xu *et al.* (2009), Chen *et al.* (2016)). Nevertheless, if the pulse-taking method, pulse-taking depth, and sample definition are not standardized, the features learned by the model may be affected by variations in measurement conditions, thereby reducing the stability and reliability of the analytical results (Chung *et al.* (2013), Chung *et al.* (2014)). Therefore, how to establish a more consistent digital analysis framework while preserving the observational logic of TCM pulse diagnosis remains a core issue in pulse diagnosis objectification research.

With the development of pulse waveform signal analysis techniques and deep learning methods, pulse waveform data have gradually been applied to various recognition tasks, including disease identification and pulse pattern classification (Zhang *et al.* (2018), Masoumi Shahrbabak *et al.* (2024), Fan *et al.* (2023), Chen *et al.* (2022)). Among these studies, Chen *et al.* (2022) proposed a method that combines Pulse-Line Intersection feature extraction with artificial intelligence-based classification for

hesitant pulse wave classification, demonstrating the feasibility of pulse waveform image analysis and model learning for choppy-pulse-related classification tasks. However, most existing studies still treat the pulse waveform as a single holistic input and rarely further investigate the differences in local detail, main contour, and global context presented by the same single pulse waveform across different field-of-view levels. For the recognition of CP and NCP, if a model relies only on a single-scale input, it may preserve only part of the discriminative information and may therefore fail to fully capture the morphological characteristics of complex pulse patterns.

In the field of medical image classification, multi-scale feature fusion has been shown to facilitate the integration of local and global information and to improve classification performance as well as the stability of feature representation (Huo *et al.* (2023), Zhu *et al.* (2024)). The same perspective has gradually extended to pulse diagnosis and pulse waveform analysis research. Existing studies have pointed out that wrist pulse recognition tasks are well suited to fusion-based representations and that features from different sources or different levels may also have complementary value (Zhang *et al.* (2021), Guo *et al.* (2022)). Therefore, if the concept of multi-scale feature fusion is introduced into two-dimensional pulse waveform image analysis, it may become possible to simultaneously preserve local waveform detail, main contour structure, and overall pulse context, thereby establishing a more complete basis for classification. This direction is also more consistent with the multi-level observational logic of TCM pulse diagnosis, which emphasizes local detail, main contour, and overall pulse context.

In addition to the integration of multi-scale information, the choice of backbone model also directly affects the efficiency and stability of feature learning. ConvNeXt, proposed by Liu *et al.* (2022), revisits the design of modern Convolutional Neural Networks (CNNs) and shows that convolution-based architectures can still maintain strong performance in image classification tasks after incorporating modern design principles. However, for binary medical image classification under small-sample conditions, model design must still maintain a balance between representational capacity and the risk of overfitting, and the network scale should therefore not be expanded without restraint (Zerouaoui and Idri (2022), Shamshiri *et al.* (2023)). On the other hand, if the data partitioning strategy is inappropriate, model performance may be overestimated due to data leakage. Accordingly, a patient-wise fixed split is particularly important in cross-subject classification studies (Veetil *et al.* (2024), Del Pup *et al.* (2025), Fan Ouyang *et al.* (2021)). In other words, for CP image classification, if the goal is to handle multi-scale information, maintain model trainability, and improve evaluation credibility at the same time, both a lightweight architectural design and a rigorous patient-wise fixed split strategy are required.

Based on the above background and research gap, this study proposes a Light ConvNeXt-style architecture based on multi-scale feature fusion for binary classification of CP and NCP. Unlike size-search-oriented studies, this work does not perform an exhaustive size search. Instead, it adopts three representative scales, 33×33, 61×61, and 105×105, corresponding to local detail, main contour, and global context, respectively, and establishes six fixed experimental settings to systematically compare single-scale models and multi-scale fusion models. Through this design, this study aims to investigate whether information from different scales provides complementary benefits and to verify the feasibility of a Light ConvNeXt-style multi-branch design under small-sample conditions with a patient-wise fixed split strategy. In addition, this study further evaluates whether such a method can better reflect the multi-level observational logic of TCM pulse diagnosis and serve as one of the technical foundations for a choppy pulse-assisted recognition system.

The remainder of this paper is organized as follows. Chapter 2 reviews the literature on pulse diagnosis objectification, pulse waveform analysis, multi-scale feature fusion, the ConvNeXt architecture, and data partitioning and performance evaluation. Chapter 3 describes the research design, including the study samples, multi-scale data construction, the Light ConvNeXt-style multi-branch model architecture, the classification procedure, and the experimental settings. Chapter 4 presents the experimental results and discussion, including the overall comparison of the six fixed experiments and the analysis of the best model and the representative control model. Chapter 5 concludes the study and

outlines future work by summarizing the main findings, research significance, study limitations, and possible directions for further extension.

2 Related Works

TCM pulse diagnosis serves as a vital foundation for syndrome differentiation and treatment planning. However, because its interpretation has historically relied on the clinical experience and subjective judgment of TCM practitioners, its objectification and standardization remain a critical research focus. Tang *et al.* (2012) proposed an artificial neural network-based TCM pulse diagnostic model, verifying its feasibility in quantifying pulse diagnosis. Shu and Sun (2017) further developed classification indices for pulse diagnosis, demonstrating that pulse information can be systematically organized into consistent quantitative descriptors. Furthermore, Xu *et al.* (2009) utilized a fuzzy neural network to show that pulse manifestations can be transformed into image representations with classification value. In addition, Chen *et al.* (2016) further analyzed arterial pulse waves using image processing and feature extraction techniques. Their study segmented measured pulse waves into arterial pulse wave curves and quantified relationships among cycle periods, slopes, and characteristic points, indicating that pulse-wave image segmentation, cycle extraction, and quantitative feature analysis can serve as an important technical basis for the objectification of pulse diagnosis. Beyond pulse digitization, the standardization of measurement conditions is a crucial prerequisite for automated recognition. Chung *et al.* (2012) noted that lacking a consistent pulse-taking standard compromises the objectivity of feature learning. Accordingly, Chung *et al.* (2015) investigated the relationship between arterial width and pulse-taking depth, highlighting the necessity for consistency in sample definition and measurement protocols.

Driven by technological advancements, pulse wave signals have been widely applied in disease recognition tasks. Zhang *et al.* (2012) demonstrated that wrist pulse signals can effectively identify diseases such as lung cancer. Similarly, Cui and Song (2022) successfully classified coronary heart disease using acoustic pulse waves. Regarding deep learning applications, Masoumi Shahrbabak *et al.* (2013) and Mair *et al.* (2022) revealed the strong potential of combining deep learning with pulse wave data for diagnosing cardiovascular diseases and arterial stenosis. Furthermore, Ouyang *et al.* (2020) utilized pulse waveform fitting techniques for diabetes detection; Fan *et al*. (2023) explored the diagnostic value of graph features across pulse cycles; and Sen *et al.* (2025) applied machine learning to analyze the pulse wave topology of virtual subjects. Notably, Chen *et al.* (2022) further proposed a hesitant pulse wave classification approach that integrates Pulse-Line Intersection-based feature extraction with artificial intelligence models. Using data from 46 patients, six measurement points, and six-second recordings, their study classified hesitant and moderate pulses, indicating that the integration of pulse-wave image analysis, feature engineering, and AI models can already be applied to hesitant-pulse-related classification tasks. While these studies collectively confirm the immense value of pulse wave data in clinical computer-aided diagnosis, existing methods typically treat the pulse wave as a single holistic input, leaving in-depth explorations of multi-scale waveform variations within two-dimensional (2D) images relatively scarce.

To overcome the limitations of relying on a single holistic input, model designs in medical image classification have increasingly emphasized the importance of multi-scale image information. Huo *et al.* (2023) pointed out that local and global features in medical images are highly complementary, and their hierarchical fusion can effectively improve classification performance. In a skin lesion classification task, Hu *et al.* (2024) demonstrated that integrating shallow and deep multi-scale features significantly enhances recognition capabilities. Zhu *et al.* (2024) further verified that multi-scale feature fusion provides stable feature representation advantages, even in class-imbalanced scenarios. Together, these studies support the premise that the core value of multi-scale design lies in integrating information

from diverse hierarchical levels to establish a more robust classification foundation. Extending the concepts of multi-scale feature fusion to the pulse diagnosis domain, Qi Zhang *et al.* (2021) proposed a multichannel feature fusion approach based on graph networks, demonstrating that wrist pulse recognition tasks are well-suited for modeling via fusion representations. Guo *et al.* (2022) emphasized the complementary value of diverse feature sources, pointing out that a single feature is often insufficient to capture the full discriminative information embedded in pulse wave signals. The aforementioned literature indicates that pulse wave analysis tasks necessitate the fusion of information from different sources or hierarchical levels to achieve more discriminative representations. This theoretical background justifies the exploration of integrating multi-scale perspectives, including local details, main contours, and global contexts, in pulse wave image analysis.

Beyond the integration of multi-scale features, the selection of an appropriate backbone network directly determines a model's effectiveness in learning local and global information. Liu *et al.* (2022) re-examined modern convolutional architecture designs and proposed ConvNeXt, demonstrating that convolutional networks, when updated with modern design principles, can maintain a strong competitive edge in image classification tasks. This establishes a design paradigm that can be easily extended and adapted for lightweight applications. In the context of medical image classification, Aruk *et al.* (2025) confirmed that ConvNeXt-based architectures serve as effective backbone networks for skin lesion classification. Gulsoy and Kablan (2025) further utilized ConvNeXt as an adaptable component for lung cancer classification. Jin *et al.* (2023) also effectively improved the classification performance of chest X-ray images by employing a pre-trained ConvNeXt as a feature extractor. These studies collectively illustrate that ConvNeXt and its variant architectures possess high applicability in medical classification tasks.

In binary medical image classification tasks with limited annotated data, the model design must strike a delicate balance between representation capacity and the risk of overfitting. Zerouaoui and Idri (2022) found that model architecture design directly impacts the stability and final performance of binary classification for breast cancer medical images. Shamshiri *et al.* (2023) similarly pointed out that when annotated data is scarce, model design and knowledge transfer strategies significantly influence recognition capabilities. Therefore, under small-sample conditions, unrestrictedly expanding the network scale is inadvisable; instead, adopting an appropriately lightweight convolutional architecture combined with a multi-branch design presents a sound methodological basis.

Beyond model design, data partitioning and performance evaluation strategies are critical for ensuring the credibility of medical classification research. Veetil *et al.* (2024) noted that improper data partitioning can easily lead to overestimated model performance due to data leakage, thereby undermining its actual generalizability. Del Pup *et al.* (2025) showed that different data partitioning strategies significantly affect the outcomes of cross-subject analyses, underscoring the necessity of subject-wise partitioning. In pulse wave signal analysis, Ouyang *et al.* (2021) directly adopted an inter-patient classification design, proving the importance of individual-level data partitioning in biomedical signal research. Furthermore, the previously mentioned study by Masoumi Shahrbabak *et al.* (2024) also employed multiple evaluation metrics, including the area under the curve (AUC), sensitivity, and specificity, reflecting the importance of comprehensive performance assessment in clinical classification tasks.

3 Lightweight Convnext-Style Multi-Scale CP Classification Method

This study proposes a lightweight ConvNeXt-style multi-scale CP classification method to compare the effects of single-scale input and multi-scale feature fusion on classification performance. The overall workflow of this study includes data acquisition and sample construction, representative multi-scale

image generation, patient-wise fixed data splitting, multi-branch Light ConvNeXt-style model construction, and classification performance evaluation, as shown in Figure 1. Algorithm 1 further summarizes the complete forward processing flow from multi-scale input to binary classification output.

The design of this study has two core objectives. First, this study constructs single-scale, dual-scale, and triple-scale input settings based on three representative scales in order to investigate the complementary effects of different scale information. Second, this study assigns an independent Light ConvNeXt-style backbone to each scale branch. All scale branches share the same architecture, but no weights are shared across branches. This design allows the model to separately learn local details, main contours, and global pulse-wave context, and then improve CP discrimination through feature fusion. Based on these objectives, this chapter is divided into four parts, including research samples and multi-scale data construction, multi-scale Light ConvNeXt-style classifier design, multi-scale CP classification procedure, and experimental design with evaluation protocol.

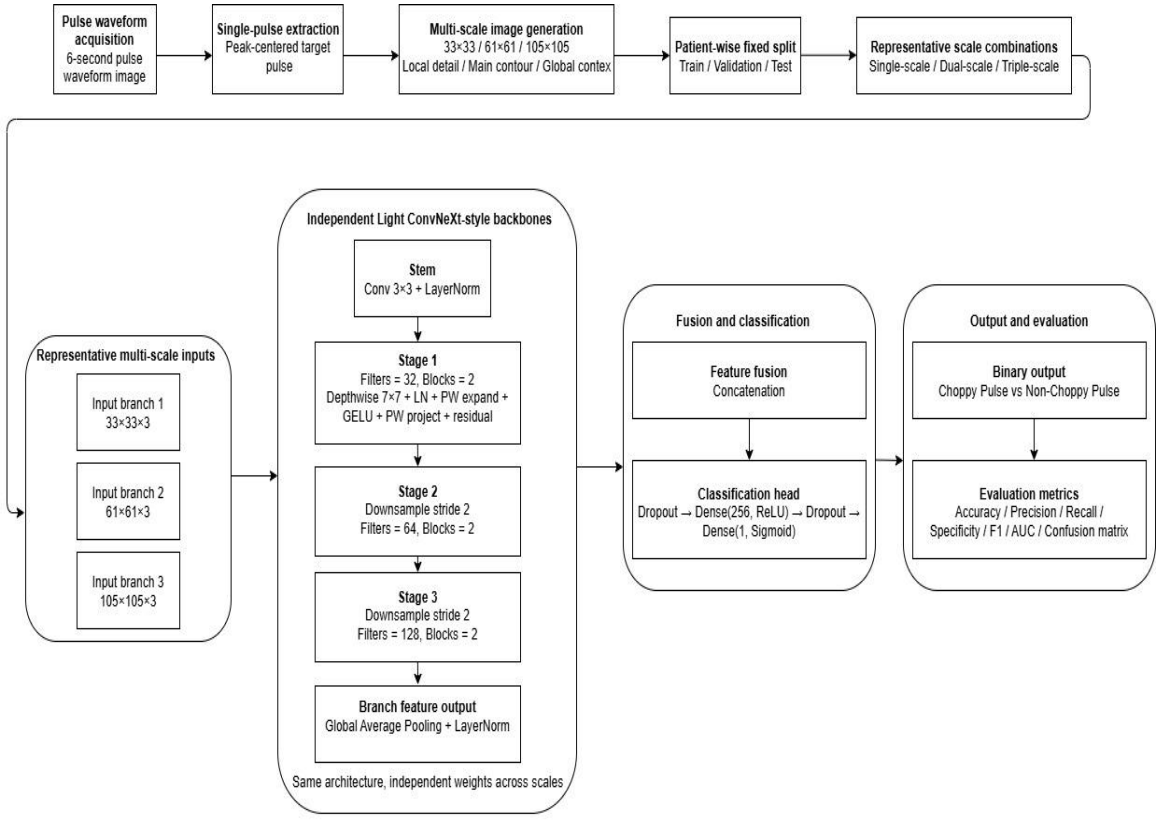


Figure 1: Integrated research workflow and model architecture of the proposed lightweight ConvNeXt-style multi-scale CP classification method.

3.1 Research Samples and Multi-Scale Data Construction

The original data used in this study were 6-second pulse waveform images acquired by a pulse diagnosis instrument. Data preprocessing includes two main steps. First, a single-pulse sample is extracted from each original pulse waveform image, and the pulse apex is used as the center reference

for localization. Second, multiple pulse waveform images at different scales are cropped from the same center position and are used as subsequent model inputs. This design allows images at different scales to correspond to the same analytical sample and helps preserve multi-level pulse-wave information.

This study uses pulse waveform images as model inputs and CP and NCP as the corresponding binary classification labels. The CP and NCP labels were established according to physician-based pulse diagnosis annotation. Pulse waveform images obtained after single-pulse extraction and multi-scale cropping were used as the model inputs, whereas CP and NCP served as the corresponding ground-truth labels.

For multi-scale image construction, this study selects three representative scales, namely 33×33, 61×61, and 105×105, for analysis. These three scales correspond to different levels of pulse-wave information. Specifically, 33×33 emphasizes local details, 61×61 represents the main contour, and 105×105 preserves a more complete global pulse-wave context. Based on this design, this study establishes representative single-scale, dual-scale, and triple-scale input settings to compare the classification effects of different scale combinations. This study further constructs three representative cropping scales centered on the waveform peak of a single pulse so as to correspond to different waveform field-of-view levels ranging from local detail to main contour and global context, as shown in Figure 2. Specifically, the 33×33 scale mainly preserves local details around the peak and is more suitable for observing subtle waveform variations; the 61×61 scale further covers the main contour of a single pulse and helps represent the core waveform morphology; and the 105×105 scale preserves a broader global context and reflects a more complete pulse-wave pattern. This design enables this study to compare the performance differences between single-scale and multi-scale fusion settings from the three levels of local detail, main contour, and global context, and to further investigate whether information from different scales provides complementary benefits.

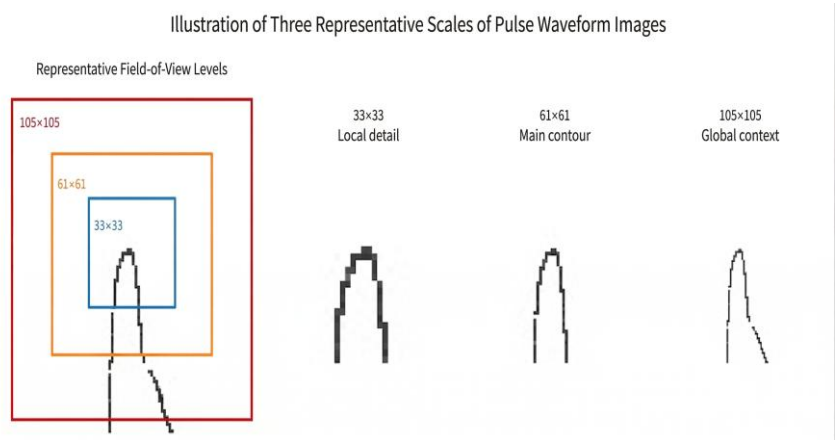


Figure 2: Illustration of the three representative pulse waveform image scales and their corresponding waveform field-of-view levels.

For data splitting, this study established a binary classification dataset consisting of CP and NCP according to physician-based pulse diagnosis annotation, and applied a patient-wise fixed split strategy to divide the data into training, validation, and test sets with a ratio of 7:2:1. Since the three representative scales, namely 33×33, 61×61, and 105×105, were all cropped from the same set of single-pulse samples, the samples across different scales maintained one-to-one correspondence. This design ensured that samples from the same subject did not appear across different subsets, thereby reducing the risk of data leakage and improving the reliability of model evaluation. The numbers of cases and images in each data subset are summarized in Table 1.

After data splitting, this study further defined six fixed experimental settings, including three single-scale models, two dual-scale fusion models, and one triple-scale fusion model, in order to systematically compare the performance differences between single-scale and multi-scale fusion settings.

Table 1: Distribution of cases and images in the CP/NCP dataset across the training, validation, and test sets

Data split	CP cases	CP images	NCP cases	NCP images	Total images
Training set (70%)	39	1,610	23	1,136	2,746
Validation set (20%)	11	441	7	329	770
Test set (10%)	6	264	3	135	399

3.2 Multi-Scale Light ConvNeXt-Style Classifier Design

This study proposes a multi-branch lightweight ConvNeXt-style classifier to process pulse waveform images at different representative scales. The overall model consists of four components, namely a representative multi-scale input module, independent Light ConvNeXt-style backbones, a feature fusion module, and a classification head. Liu *et al*. (2022) proposed ConvNeXt as a modernized convolutional architecture that reorganizes the design of contemporary convolutional neural networks while preserving the advantage of convolution in local feature extraction. Based on this design concept, this study further redesigns a lightweight ConvNeXt-style backbone according to the requirements of multi-scale CP image classification. Since the target task of this study is small-sample CP image classification and the model must simultaneously support multi-branch and multi-scale feature fusion, this study does not directly adopt the full official ConvNeXt backbone. Instead, this study retains its core block design elements and reconstructs a more lightweight backbone to reduce model complexity and improve training stability, reproducibility, and practical feasibility.

At the input side, the model receives single-scale, dual-scale, or triple-scale images according to the experimental setting. When multiple scales are used, each scale image is fed into its corresponding branch separately. Each branch accepts one fixed input size in three-channel image form in order to preserve scale-specific spatial information.

For backbone design, each input branch is assigned an independent Light ConvNeXt-style backbone. All scale branches share the same architecture, but no weights are shared across branches. This design allows each scale input to learn its most suitable feature representation independently without being constrained by parameter sharing across different scales. Each backbone consists of a stem module, three feature extraction stages, and a branch feature output layer. The stem module contains a 3×3 convolution layer and a Layer Normalization layer for generating the initial feature representation. The model then sequentially enters Stage 1, Stage 2, and Stage 3. Stage 1 uses 32 filters and two lightweight ConvNeXt-style blocks for feature extraction. Stage 2 and Stage 3 apply downsampling (stride = 2) followed by 64 and 128 filters, respectively, to learn higher-level pulse-wave features. At the end of each branch, Global Average Pooling and Layer Normalization are applied to generate the branch feature vector for subsequent multi-scale fusion.

For feature fusion, this study uses concatenation as the default fusion method. In the single-scale setting, the feature vector from the selected branch is directly used as the input of the classification head. In the multi-scale setting, the feature vectors from all selected branches are concatenated along the feature dimension to form the fused multi-scale representation, and the classification head then learns the relationships among different scales.

For the classification head, the fused feature vector is sequentially passed through Dropout, Dense (256, ReLU), Dropout, and Dense (1, Sigmoid). The two Dropout layers reduce the risk of overfitting, the fully connected layer performs nonlinear feature transformation, and the final Sigmoid output layer generates the predicted probability that the sample belongs to Choppy Pulse. The model then converts this probability into the final binary classification result according to a predefined classification threshold.

3.3 Multi-Scale CP Classification Procedure

To more clearly describe the complete forward processing flow of the proposed lightweight ConvNeXt-style multi-scale classification model, including multi-scale image input, branch-wise feature extraction, feature fusion, binary prediction, and performance evaluation, this study further summarizes the procedure as *Algorithm 1*.

Algorithm 1. Proposed Lightweight ConvNeXt-Style Multi-Scale CP Classification Procedure.
Input: *A set of aligned multi-scale pulse waveform images corresponding to one sample, with representative input scales selected from {33×33, 61×61, 105×105}.*
Output: *The predicted probability p that the input sample belongs to CP, and the final binary classification result $\hat{y}$.*

Begin

S*tep 1: Read the aligned pulse waveform image paths corresponding to the selected representative scale combination.*

Step 2: For each selected input scale $s \in \{33, 61, 105\}$, read the corresponding image file.

Step 3: Check the number of image channels. If the input image is a grayscale image, convert it into a three-channel RGB image. If the input image contains more than three channels, retain the first three channels as the model input.

Step 4: Resize each input image to the specified spatial size $s \times s \times 3$.

Step 5: Normalize the resized image by converting pixel values into floating-point format and scaling them to the range [0,1].

Step 6: Feed each preprocessed image branch into its corresponding independent Light ConvNeXt-style backbone.

Step 7: In each backbone, apply the stem module composed of a 3×3 convolution layer and a Layer Normalization layer to generate the initial feature representation.

Step 8: Feed the feature map into Stage 1, where two lightweight ConvNeXt-style blocks with 32 filters are applied. Each block consists of a depthwise 7×7 convolution, Layer Normalization, pointwise expansion, GELU activation, pointwise projection, and residual addition.

Step 9: Feed the resulting feature map into Stage 2. First perform downsampling with stride 2, and then apply two lightweight ConvNeXt-style blocks with 64 filters.

Step 10: Feed the resulting feature map into Stage 3. First perform downsampling with stride 2, and then apply two lightweight ConvNeXt-style blocks with 128 filters.

Step 11: Apply Global Average Pooling and Layer Normalization to the output of each branch to obtain the branch-level feature vector.

Step 12: If only one input scale is selected, directly use the corresponding branch feature vector as the fused representation. Otherwise, concatenate the feature vectors from all selected branches along the feature dimension to form the fused multi-scale representation.

Step 13: Feed the fused representation into the classification head.

Step 14: Apply a Dropout layer to the fused feature vector to reduce the risk of overfitting.

Step 15: Feed the processed feature vector into a fully connected layer with 256 neurons and a ReLU activation function for nonlinear feature transformation.

Step 16: Apply a second Dropout layer to the transformed feature vector.

Step 17: Feed the resulting representation into a final fully connected layer with one neuron and a Sigmoid activation function to generate the predicted probability p that the input sample belongs to CP.

Step 18: Convert the probability value p into the final binary classification result $\hat{y}$ according to the predefined classification threshold.

Step 19: For model evaluation, compare the predicted results with the ground-truth labels and compute Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC, and the Confusion Matrix.

End.

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The input of the algorithm is a set of aligned multi-scale pulse waveform images, where the scales are selected from the representative scale set of 33×33, 61×61, and 105×105. The algorithm first reads the images at the selected scales and then sequentially performs channel checking, three-channel conversion, resizing, and pixel normalization. After preprocessing, each scale image is fed into its corresponding independent Light ConvNeXt-style backbone for feature extraction. In the single-scale setting, the branch feature vector is directly passed to the classification head. In the multi-scale setting, the feature vectors from different branches are first concatenated and then passed to the classification head to generate the predicted probability p that the sample belongs to CP. Finally, the model converts p into the final binary classification result $\hat{y}$ according to a predefined threshold and computes the performance evaluation metrics.

It should be further noted that this study defines the positive class as CP and the negative class as NCP. This definition directly affects the computation and interpretation of the confusion matrix as well as Recall, Specificity, Precision, and F1-score. Therefore, all subsequent evaluations in this study follow this class definition.

3.4 Experimental Design and Evaluation Protocol

The experimental design of this study aims to compare the classification performance and computational efficiency between single-scale models and multi-scale fusion models. Instead of performing exhaustive scale searching, this study establishes six fixed experimental settings based on three representative scales and their combinations. These include single-scale inputs of ConvNeXt-33, ConvNeXt-61, and ConvNeXt-105, dual-scale fusion settings of ConvNeXt-33+61 and ConvNeXt-61+105, and a triple-scale fusion setting of ConvNeXt-33+61+105. This fixed design enables systematic analysis of the complementary effects between single-scale and multi-scale inputs.

For training configuration, this study uses binary cross-entropy as the loss function and Adam as the optimizer. The training process is combined with EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint to preserve the model weights with the best validation performance. This study adopts the best validation AUC as the primary criterion for model selection. When multiple models achieve similar performance on the primary metric, additional ranking is conducted according to best validation accuracy, test-set F1-score, and training time.

For performance evaluation, this study calculates Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score, AUC, and the Confusion Matrix on the test set. Accuracy measures the overall correctness of classification. Precision measures the proportion of actual CP samples among those predicted as CP. Recall (Sensitivity) measures the proportion of correctly identified CP samples among all actual CP samples. Specificity measures the proportion of correctly identified NCP samples among all actual NCP samples. F1-score reflects the balance between Precision and Recall, while AUC

evaluates the overall discriminative ability of the classifier. In addition, this study reports the learning curves, ROC curves, and confusion matrices of the best model and the representative control model to support a more complete interpretation of the results.

4 Experimental Results and Discussion

To compare the performance differences between single-scale models and multi-scale fusion models, this study conducted a systematic analysis of six fixed experimental settings, using best validation AUC, test F1-score, and training time as the three primary comparison dimensions. The overall results show that the models did not follow a fully consistent ranking across validation performance, balanced classification performance on the test set, and training cost. Therefore, the focus of this chapter is not merely to identify a single top-scoring model, but to clarify the distinct roles and complementary relationships of information at different scales in cp image classification. The overall comparison of the six fixed experiments is presented in Table 2. In terms of the primary selection criterion, best validation AUC, ConvNeXt-33+61+105 achieved the highest value of 97.13%, making it the best validation model in the main experiment. ConvNeXt-33 also reached 97.09%, showing only a minimal difference from the three-scale model, while ConvNeXt-33+61 maintained a comparably high value of 96.59%. In contrast, ConvNeXt-105 achieved only 50.45%, which was markedly lower than the other settings. These results indicate that three-scale fusion can improve peak validation performance in the main experiment, but its advantage does not substantially separate it from all other models. In particular, the 33×33 small-scale model already demonstrated highly competitive discriminative performance, as shown in Figure 3.

Table 2: Overall comparison of the six fixed experimental settings.

Experiment	Model Type	Best validation AUC	Best validation accuracy	Test accuracy	Test F1-score	Test AUC	Training time (second)
ConvNeXt-33	Single-scale	97.09%	92.47%	91.23%	93.23%	95.06%	96.11
ConvNeXt-61	Single-scale	95.16%	91.56%	91.98%	94.03%	93.96%	198.66
ConvNeXt-105	Single-scale	50.45%	57.27%	66.17%	79.64%	51.13%	147.12
ConvNeXt-33+61	Dual-scale fusion	96.59%	92.99%	88.47%	90.91%	93.79%	162.29
ConvNeXt-61+105	Dual-scale fusion	95.09%	90.78%	92.48%	94.51%	95.44%	884.26
ConvNeXt-33+61+105	Triple-scale fusion	97.13%	93.25%	90.48%	92.64%	94.10%	964.33

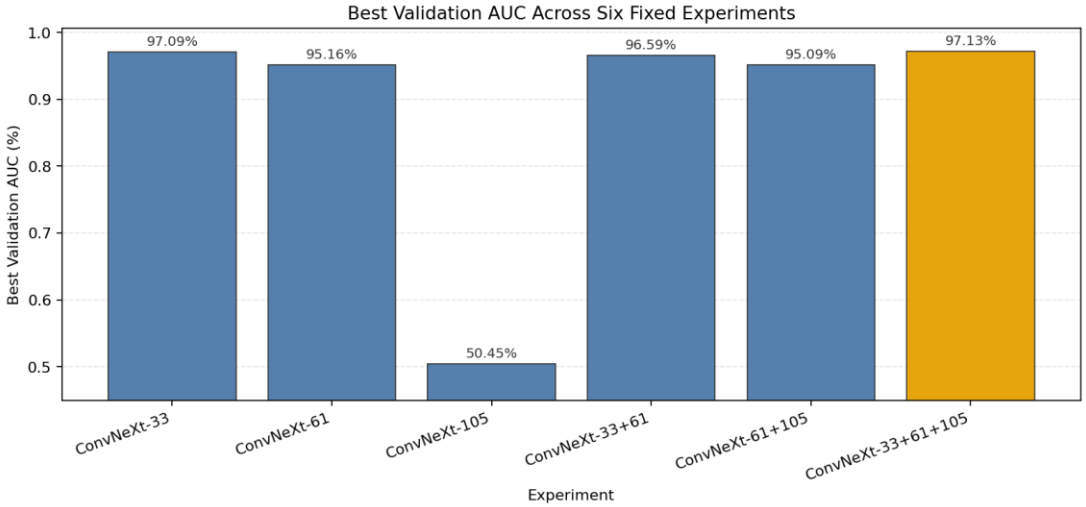


Figure 3: Comparison of best validation AUC across six fixed experiments.

When balanced classification performance on the test set is further considered, the ranking differs from that observed on the validation set. ConvNeXt-61+105 achieved the highest test F1-score of 94.51%, making it the best-performing model on the test set. ConvNeXt-61 and ConvNeXt-33 followed closely at 94.03% and 93.23%, respectively, and both remained highly competitive. ConvNeXt-33+61+105 achieved a test F1-score of 92.64%, which was still stable but not the highest under this metric, whereas ConvNeXt-105 dropped to 79.64%. These results suggest that multi-scale fusion can indeed provide complementary benefits on the test set, but the best-performing combination is not necessarily the one with the largest number of branches. In particular, the combination of 61×61 and 105×105 achieved stronger balanced classification performance, indicating meaningful complementarity between the main contour and the global context, as shown in Figure 4. In terms of training cost, ConvNeXt-33 required only 96.11 s and was the most efficient among all six settings, whereas ConvNeXt-61+105 and ConvNeXt-33+61+105 required 884.26 s and 964.33 s, respectively. This finding shows that higher-level multi-branch fusion may improve performance, but it also incurs substantially higher computational cost, as shown in Figure 5. Taken together, these three comparisons do not support a one-directional conclusion that multi-scale models must always outperform single-scale models. Instead, the overall pattern is more consistent with a “small-scale dominant, multi-scale complementary” interpretation.

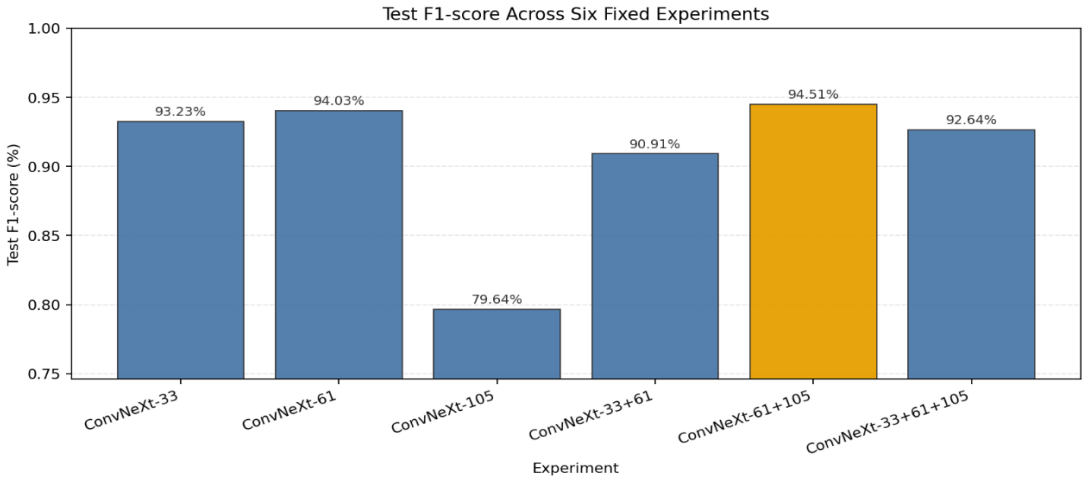


Figure 4: Comparison of test F1-score across six fixed experiments.

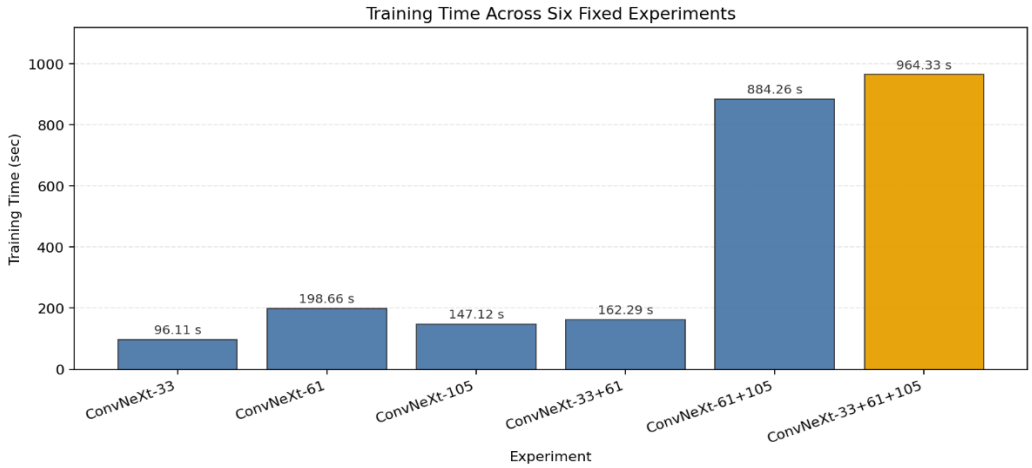


Figure 5: Comparison of training time across six fixed experiments.

Based on the model selection criterion adopted in this study, ConvNeXt-33+61+105 was selected as the best model in the main experiment because it achieved the highest best validation AUC. The learning curves show that both train_acc and val_acc rose rapidly in the early stage and then remained stable at a high level, while train_loss and val_loss decreased quickly before gradually flattening. No obvious sign of overfitting was observed overall, indicating that the three-scale fusion model maintained stable convergence behavior in this task, as shown in Figure 6(a)-(b). The ROC curve further shows a test AUC of 0.9410, with the curve remaining clearly above the random baseline, indicating strong overall discriminative ability between CP and NCP, as shown in Figure 7(a). The confusion matrix shows that the model correctly classified the majority of CP and NCP samples, with relatively similar Sensitivity and Specificity and an overall balanced error distribution. However, some CP samples were still misclassified, suggesting that certain CP samples may still share ambiguous boundaries or overlapping representations with NCP samples, as shown in Figure 8(a). These results indicate that integrating local detail, main contour, and global context can form a more complete discriminative

representation, and they support the feasibility of applying a Light ConvNeXt-style multi-branch design to CP image classification.

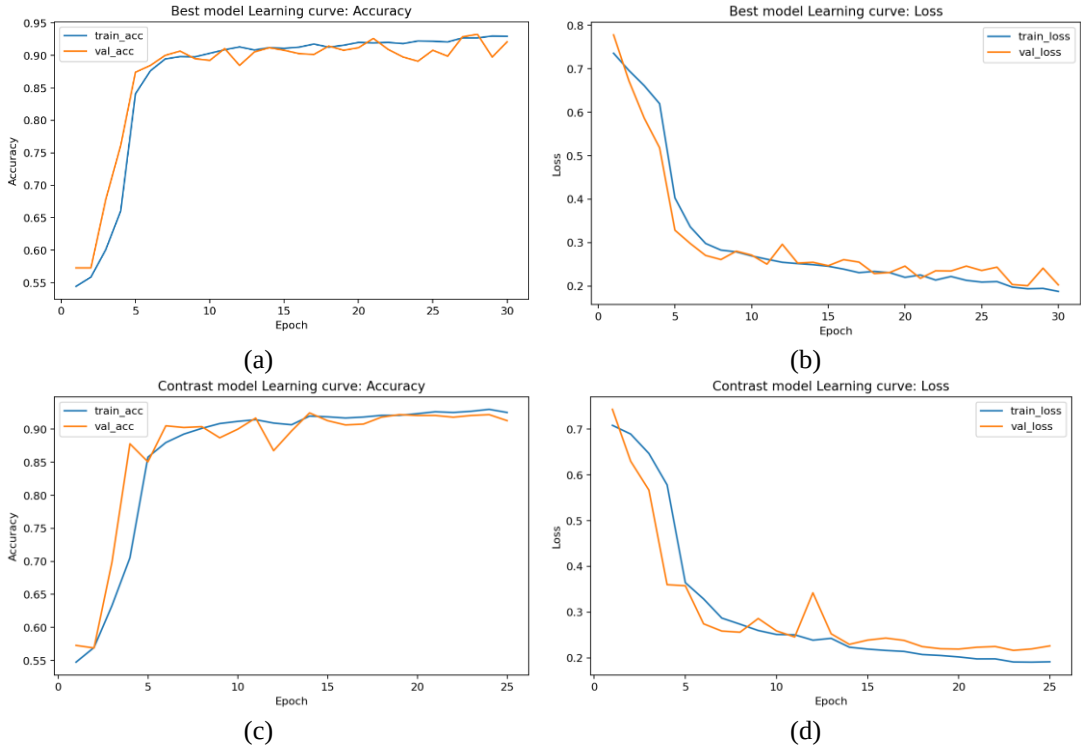
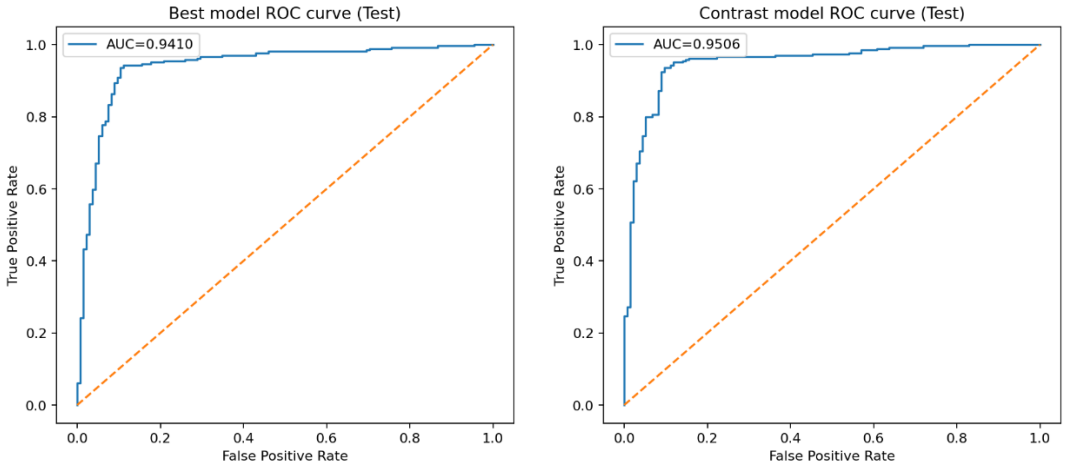
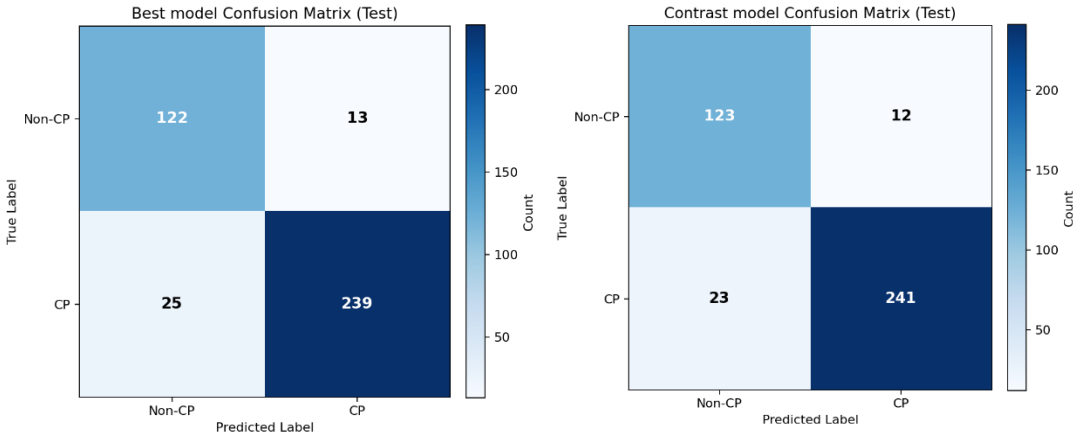


Figure 6: Learning curves of the representative model and the control model: (a) accuracy curve of ConvNeXt-33+61+105; (b) loss curve of ConvNeXt-33+61+105; (c) accuracy curve of ConvNeXt-33; and (d) loss curve of ConvNeXt-33.



(a) (b)

Figure 7: ROC curves of the representative model and the control model: (a) test ROC curve of ConvNeXt-33+61+105; (b) test ROC curve of ConvNeXt-33.



(a) (b)

Figure 8: Confusion matrices of the representative model and the control model: (a) test confusion matrix of ConvNeXt-33+61+105; (b) test confusion matrix of ConvNeXt-33.

To further clarify the difference between single-scale learning and multi-scale fusion, this study selected ConvNeXt-33 as the representative control model. This model was not only the strongest among the single-scale settings, but also the fastest among all six fixed experiments, and therefore serves as a representative baseline for the local-detail scale. The results show that ConvNeXt-33 achieved a best validation AUC of 97.09%, which was very close to that of the best three-scale model, and its test ROC AUC reached 0.9506, which was slightly higher than that of ConvNeXt-33+61+105. Further comparison of the learning curves, ROC curve, and confusion matrix shows that ConvNeXt-33

maintained performance similar to that of the best three-scale model in terms of convergence stability, discriminative ability, and error distribution, as shown in Figure 6(c)-(d), Figure 7(b), and Figure 8(b). The confusion matrix further indicates that ConvNeXt-33 produced slightly fewer overall classification errors than the best three-scale model, suggesting a good balance between Sensitivity and Specificity. These findings indicate that the local detail captured by the 33×33 scale already contains highly informative discriminative cues for CP recognition. In other words, local waveform detail is not merely a secondary supplement, but one of the core foundations supporting classification performance. By contrast, the value of multi-scale fusion does not lie in fully replacing single-scale learning, but in supplementing contour-level and contextual information under specific combinations so that the model can achieve more complete performance across different evaluation criteria.

Taken together, the main implications of this chapter can be summarized in three points. First, the discriminative information in CP images exhibits a clear multi-scale hierarchical structure. ConvNeXt-33 remained highly competitive in validation performance, test performance, and training efficiency, indicating that the local detail represented by the 33×33 scale serves as an important discriminative basis. Meanwhile, the fact that ConvNeXt-61+105 achieved the best test F1-score indicates that the main contour and the global context can still provide meaningful complementary information on the test set. In addition, ConvNeXt-33+61+105 achieved the highest best validation AUC, further suggesting that integrating all three field-of-view levels still has the potential to improve peak validation performance. These findings indicate that local detail, main contour, and global context in CP images are not homogeneous sources of information, but jointly contribute to classification through different scale levels. The overall comparison of the six fixed experiments is presented in Table 2, while the overall trends of the main metrics are shown in Figs. 3-5.

Second, the value of multi-scale fusion lies in complementing, rather than replacing, single-scale learning. As reviewed in Chapter 2, prior studies on multi-scale feature fusion suggest that features from different levels may be complementary. The present results further show that such complementarity must be verified under specific tasks and scale combinations, rather than assumed directly from the number of branches. In this study, the small scale provided the main discriminative source, whereas the medium and large scales supplemented contour and contextual information under particular combinations, allowing the models to achieve more complete performance across different metrics. Conversely, the clearly weaker performance of ConvNeXt-105 indicates that preserving only the global context, without sufficient support from local detail and main contour, is not enough for stable classification. Therefore, the results of this study are more appropriately interpreted through a "small-scale dominant, multi-scale complementary" framework, rather than viewing multi-scale fusion as a complete alternative to single-scale modeling.

Third, the Light ConvNeXt-style multi-branch design is feasible under a small-sample condition with a patient-wise fixed split. The learning curves, ROC curves, and confusion matrices of both the best model and the representative control model show overall stable convergence, good discriminative ability, and relatively balanced error distributions. Because this study adopted a patient-wise fixed split strategy, the resulting performance more realistically reflects the difficulty of cross-subject classification and thus makes the comparisons in this chapter more credible. Overall, the contribution of this study lies in showing that CP image classification can be understood through a multi-scale hierarchical information structure, that multi-scale fusion can serve as a complementary design strategy to improve the completeness of model performance, and that a Light ConvNeXt-style architecture can provide a practically promising classification solution under limited-data conditions. These findings further suggest that the proposed design has both methodological feasibility and practical value for the development of a choppy pulse-assisted recognition system.

5 Conclusion and future work

This study aimed to address binary classification of CP images by proposing a multi-scale feature fusion method based on a Light ConvNeXt-style architecture. Using three representative scales, 33×33, 61×61, and 105×105, six fixed experimental settings were constructed to systematically compare single-scale models with multi-scale fusion models. The results indicate that the proposed method is feasible and that information at different scales indeed plays distinct hierarchical roles in CP recognition. In the main experiment, ConvNeXt-33+61+105 achieved the highest best validation AUC, indicating that integrating local detail, main contour, and global context still has the potential to improve peak validation performance. Meanwhile, ConvNeXt-33 remained highly competitive among the single-scale models, whereas ConvNeXt-61+105 achieved the best test F1-score. Overall, the findings of this study are more consistent with a "small-scale dominant, multi-scale complementary" interpretation, rather than viewing multi-scale fusion as a complete replacement for single-scale modeling.

The significance of this study can be summarized from three perspectives. First, the discriminative information in CP images is not homogeneous, but exhibits a clear multi-scale hierarchical structure. The strong performance of the 33×33 setting indicates that local waveform detail serves as an important basis for CP classification, whereas the 61×61 main contour and the 105×105 global context can further supplement waveform information under specific combinations. Second, the value of multi-scale fusion lies in complementing discriminative information across different field-of-view levels, rather than directly replacing single-scale models with more branches. This finding is consistent with the original design objective of this study and also corresponds to the multi-level observational logic of traditional Chinese medicine pulse diagnosis, which considers local detail, main contour, and overall pulse context together. Third, this study demonstrates the feasibility of a Light ConvNeXt-style multi-branch design under a small-sample condition with a patient-wise split. As shown in Chapter 4, the models produced reasonable results in terms of convergence stability, overall discriminative ability, and balanced error distribution, suggesting that this design can serve as one of the technical foundations for a choppy pulse-assisted recognition system.

Beyond methodological feasibility, this study also has practical value. Traditional Chinese medicine pulse diagnosis has long relied on physician experience and subjective judgment. Accordingly, establishing a more consistent digital analysis process without departing from the observational logic of traditional Chinese medicine remains an important direction for the objectification of pulse diagnosis. The proposed multi-scale Light ConvNeXt-style architecture was not designed simply to pursue the highest score, but rather to construct an image classification approach that more closely reflects the multi-level observational process of pulse diagnosis through local detail, main contour, and global context. The results suggest that this direction is reasonable and further indicate that choppy pulse-assisted image recognition does not need to be restricted to a single field of view, but may benefit from appropriate scale design to improve both the completeness of model performance and the interpretability of the overall methodological framework.

Nevertheless, several limitations of this study should be noted. First, the dataset remains limited in scale, and the present task focuses on binary classification between CP and NCP. Therefore, the current conclusions should be interpreted within the data condition and task setting of this study. Second, although multi-scale fusion showed complementary benefits under some metrics, the advantages of different models were not fully consistent across all evaluation dimensions, indicating that the effectiveness of multi-scale design still depends on specific objectives. Third, although a patient-wise fixed split was adopted to improve the credibility of evaluation, the present findings are still derived from a single research framework and would benefit from further validation using broader data sources and more diverse scenarios. Finally, this study mainly focused on model performance and methodological comparison. Further work may strengthen visualization-based interpretation of model decisions and improve integration with clinical use workflows.

Based on these limitations, several directions may be considered in future work. First, the dataset size and subject sources may be expanded to further examine the stability and generalizability of the proposed framework under broader sample conditions. Second, the multi-scale design may be extended to additional pulse categories or multi-class classification tasks in order to evaluate its applicability in more complex pulse diagnosis scenarios. Third, explainable analysis methods may be incorporated to identify the key waveform regions associated with each scale branch during classification, thereby strengthening the connection between model outputs and traditional Chinese medicine pulse interpretation. Fourth, while maintaining the lightweight design, future studies may further optimize the multi-branch fusion strategy to improve the overall balance among performance, computational cost, and stability. Overall, this study provides an initial validation of the feasibility of applying multi-scale feature fusion and a Light ConvNeXt-style architecture to CP image classification, and offers a concrete foundation for subsequent methodological development and practical implementation of choppy pulse-assisted recognition systems.

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References

[1] I. Aruk, I. Pacal, and A. N. Toprak, “A novel hybrid ConvNeXt-based approach for enhanced skin lesion classification,” *Expert Systems with Applications*, vol. 283, Art. no. 127721, 2025, doi: 10.1016/j.eswa.2025.127721.

[2] H.-C. Chen, S.-S. Kuo, S.-C. Sun, and C.-H. Chang, “A Distinguishing Arterial Pulse Waves Approach by Using Image Processing and Feature Extraction Technique,” Journal of Medical Systems, vol. 40, Art. no. 215, 2016, doi: 10.1007/s10916-016-0568-4.

[3] H.-C. Chen, C. Damarjati, K. T. Putra, H.-M. Chen, C.-L. Hsieh, H.-J. Lin, M.-Y. Wu, and C.-S. Chen, “Pulse-line intersection method with unboxed artificial intelligence for hesitant pulse wave classification,” Information Processing and Management, vol. 59, Art. no. 102855, 2022, doi: 10.1016/j.ipm.2021.102855.

[4] C.-Y. Chung, Y.-W. Cheng, and C.-H. Luo, “Neural network study for standardizing pulse-taking depth by the width of artery,” Computers in Biology and Medicine, vol. 57, pp. 26–31, 2015, doi: 10.1016/j.compbiomed.2014.10.016.

[5] Y.-F. Chung, C.-S. Hu, C.-C. Yeh, and C.-H. Luo, “How to standardize the pulse-taking method of traditional Chinese medicine pulse diagnosis,” Computers in Biology and Medicine, vol. 43, no. 4, pp. 342–349, 2013, doi: 10.1016/j.compbiomed.2012.12.010.

[6] J. Cui and L. Song, “Wrist pulse diagnosis of stable coronary heart disease based on acoustics waveforms,” Computer Methods and Programs in Biomedicine, vol. 214, Art. no. 106550, 2022, doi: 10.1016/j.cmpb.2021.106550.

[7] F. Del Pup, A. Zanola, L. F. Tshimanga, A. Bertoldo, L. Finos, and M. Atzori, “The role of data partitioning on the performance of EEG-based deep learning models in supervised cross-subject analysis: A preliminary study,” Computers in Biology and Medicine, vol. 196, pt. A, Art. no. 110608, 2025, doi: 10.1016/j.compbiomed.2025.110608.

[8] L. Fan, X. Shi, Z. Wang, R. Zhang, and J. Zhang, “Disease identification method based on graph features between pulse cycles,” Biomedical Signal Processing and Control, vol. 83, Art. no. 104670, 2023, doi: 10.1016/j.bspc.2023.104670.
[9] T. Gulsoy and E. B. Kablan, “FocalNeXt: A ConvNeXt augmented FocalNet architecture for lung cancer classification from CT-scan images,” Expert Systems with Applications, vol. 261, Art. no. 125553, 2025, doi: 10.1016/j.eswa.2024.125553.
[10] C. Guo, Z. Jiang, and D. Zhang, “Multi-Feature Complementary Learning for Diabetes Mellitus Detection Using Pulse Signals,” IEEE Journal of Biomedical and Health Informatics, vol. 26, no. 11, pp. 5684–5694, 2022, doi: 10.1109/JBHI.2022.3198792.
[11] X. Huo, G. Sun, S. Tian, Y. Wang, L. Yu, J. Long, W. Zhang, and A. Li, “HiFuse: Hierarchical multi-scale feature fusion network for medical image classification,” Biomedical Signal Processing and Control, vol. 87, pt. A, Art. no. 105534, 2024, doi: 10.1016/j.bspc.2023.105534.
[12] Z. Hu, W. Mei, H. Chen, and W. Hou, “Multi-scale feature fusion and class weight loss for skin lesion classification,” Computers in Biology and Medicine, vol. 176, Art. no. 108594, 2024, doi: 10.1016/j.compbiomed.2024.108594.
[13] Y. Jin, H. Lu, W. Zhu, and W. Huo, “Deep learning based classification of multi-label chest X-ray images via dual-weighted metric loss,” Computers in Biology and Medicine, vol. 157, Art. no. 106683, 2023, doi: 10.1016/j.compbiomed.2023.106683.
[14] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 11966–11976, doi: 10.1109/CVPR52688.2022.01167.
[15] A. Mair, M. Wisotzki, and S. Bernhard, “Classification and regression of stenosis using an in-vitro pulse wave data set: Dependence on heart rate, waveform and location,” Computers in Biology and Medicine, vol. 151, pt. A, Art. no. 106224, 2022, doi: 10.1016/j.compbiomed.2022.106224.
[16] S. Masoumi Shahrbabak, S. Kim, B. D. Youn, H.-M. Cheng, C.-H. Chen, R. Mukkamala, and J.-O. Hahn, “Peripheral artery disease diagnosis based on deep learning-enabled analysis of non-invasive arterial pulse waveforms,” Computers in Biology and Medicine, vol. 168, Art. no. 107813, 2024, doi: 10.1016/j.compbiomed.2023.107813.
[17] C. Ouyang, Z. Gan, J. Zhen, Y. Guan, X. Zhu, and P. Zhou, “Inter-Patient Classification With Encoded Peripheral Pulse Series and Multi-Task Fusion CNN: Application in Type 2 Diabetes,” IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 8, pp. 3130–3140, 2021, doi: 10.1109/JBHI.2021.3061114.
[18] C. Ouyang, P. Zhou, and Z. Gan, “A non-invasive test method for type-2 diabetes mellitus by pulse waveform fitting,” Biomedical Signal Processing and Control, vol. 60, Art. no. 102000, 2020, doi: 10.1016/j.bspc.2020.102000.
[19] A. Sen, M. Aguirre, P. H. Charlton, L. Navarro, S. Avril, and J. Alastruey, “Machine learning-based pulse wave analysis for classification of circle of Willis topology: An in silico study with 30,618 virtual subjects,” Biomedical Signal Processing and Control, vol. 100, pt. B, Art. no. 106999, 2025, doi: 10.1016/j.bspc.2024.106999.
[20] M. A. Shamshiri, A. Krzyżak, M. Kowal, and J. Korbicz, “Compatible-domain Transfer Learning for Breast Cancer Classification with Limited Annotated Data,” Computers in Biology and Medicine, vol. 154, Art. no. 106575, 2023, doi: 10.1016/j.compbiomed.2023.106575.
[21] J.-J. Shu and Y. Sun, “Developing classification indices for Chinese pulse diagnosis,” Complementary Therapies in Medicine, vol.15, no.3, pp.190–198, 2007, doi: 10.1016/j.ctim.2006.06.004.
[22] A.C.Y. Tang, J. W. Y. Chung, and T. K. S. Wong, “Validation of a Novel Traditional Chinese Medicine Pulse Diagnostic Model Using an Artificial Neural Network,” Evidence-Based Complementary and Alternative Medicine, vol. 2012, Art. no. 685094, 2012, doi: 10.1155/2012/685094.

[23] I. K. Veetil, D. E. Chowdary, P. N. Chowdary, S. V., and E. A. Gopalakrishnan, “An analysis of data leakage and generalizability in MRI based classification of Parkinson's Disease using explainable 2D Convolutional Neural Networks,” Digital Signal Processing, vol. 147, Art. no. 104407, 2024, doi: 10.1016/j.dsp.2024.104407.
[24] L. Xu, M. Q.-H. Meng, K. Wang, W. Lu, and N. Li, “Pulse images recognition using fuzzy neural network,” Expert Systems with Applications, vol. 36, no. 2, pt. 2, pp. 3805–3811, 2009, doi: 10.1016/j.eswa.2008.02.028.
[25] H. Zerouaoui and A. Idri, “Deep hybrid architectures for binary classification of medical breast cancer images,” Biomedical Signal Processing and Control, vol. 71, Art. no. 103226, 2022, doi: 10.1016/j.bspc.2021.103226.
[26] Z. Zhang, Y. Zhang, L. Yao, H. Song, and A. Kos, “A sensor-based wrist pulse signal processing and lung cancer recognition,” Journal of Biomedical Informatics, vol. 79, pp. 107–116, 2018, doi: 10.1016/j.jbi.2018.01.009.
[27] Q. Zhang, J. Zhou, and B. Zhang, “Graph Based Multichannel Feature Fusion for Wrist Pulse Diagnosis,” IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 10, pp. 3732–3743, 2021, doi: 10.1109/JBHI.2020.3045274.
[28] Z. Zhu, Y. Liu, C.-A. Yuan, X. Qin, and F. Yang, “A diffusion model multi-scale feature fusion network for imbalanced medical image classification research,” Computer Methods and Programs in Biomedicine, vol. 256, Art. no. 108384, 2024, doi: 10.1016/j.cmpb.2024.108384.